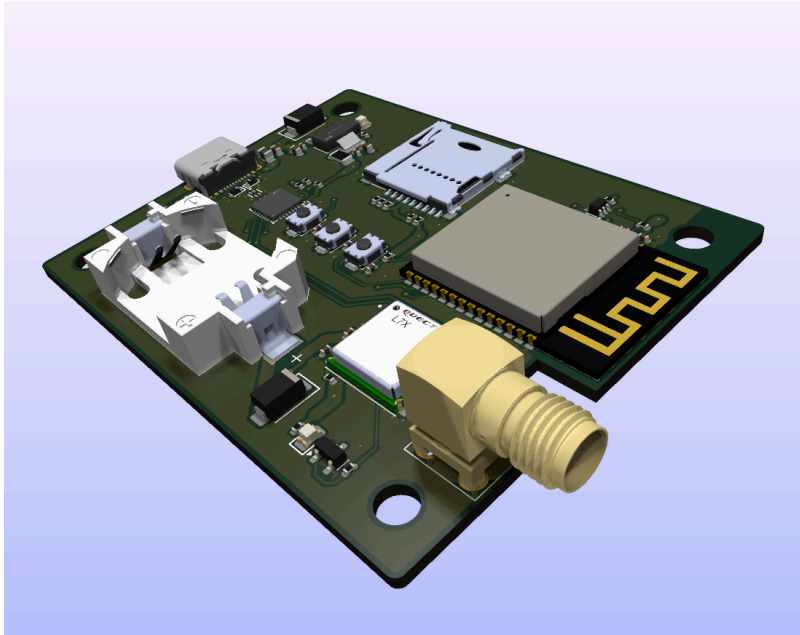


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Mileage Logger Project



Scope

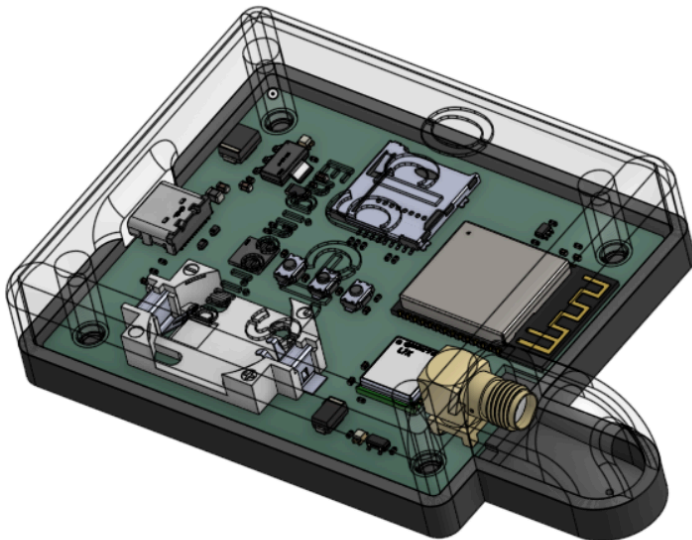
To log and display mileage for business purposes. With the ability for switching from personal journey to business journey.

Technical Design Skills

- RF waveguide design with a GCPW (Grounded Coplanar Waveguide)
- Differential pair routing for USB C datalines
- EMC and transient considerations
 - TVS (Transient Voltage Suppression)
 - Common mode choke
 - 4 layer board design with ground pours and power plane

Display

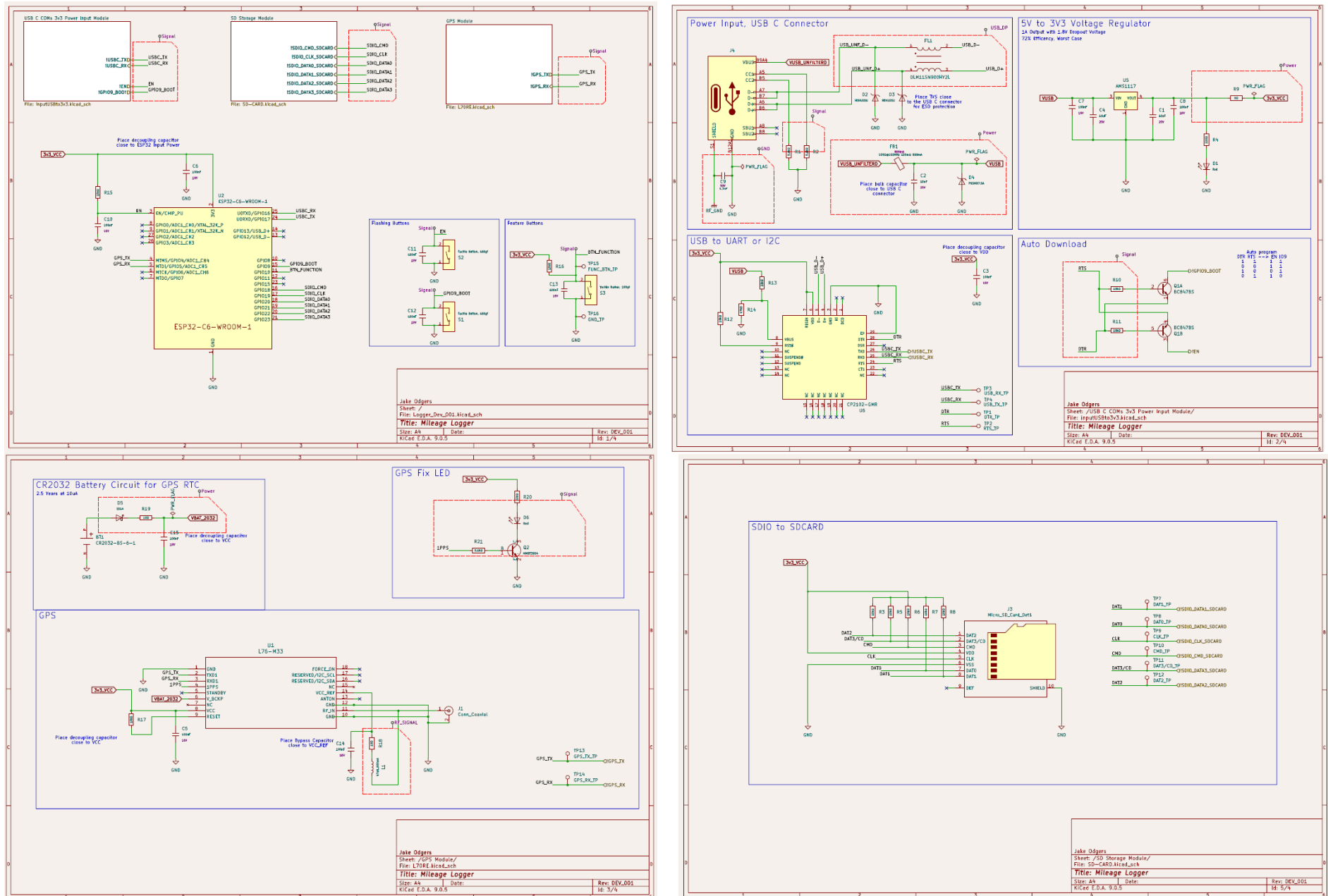
Option 1 - Display through a GUI on a computer made through Python Tkinter.



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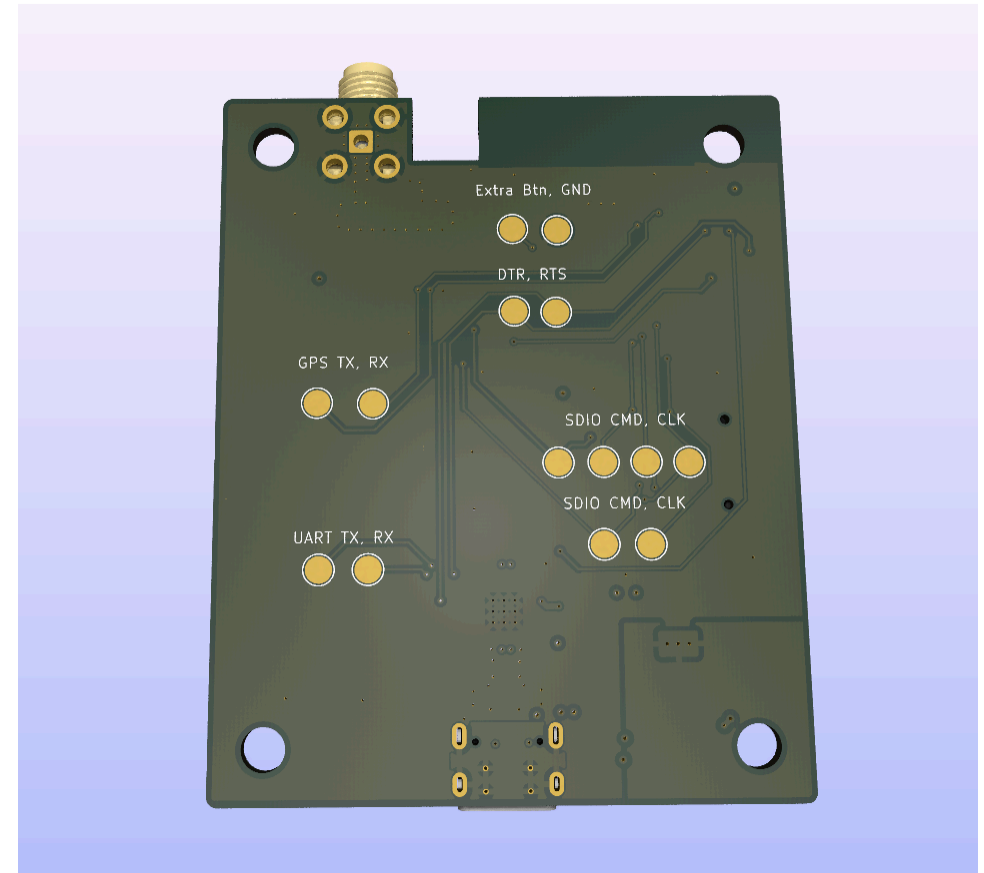
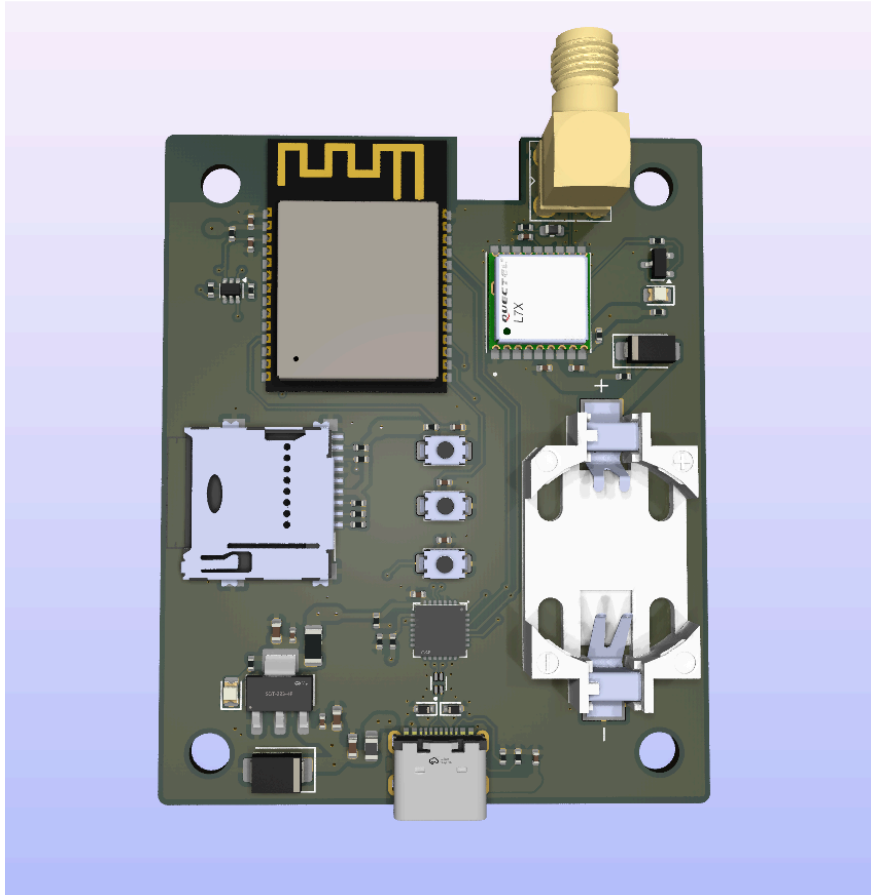
Schematic



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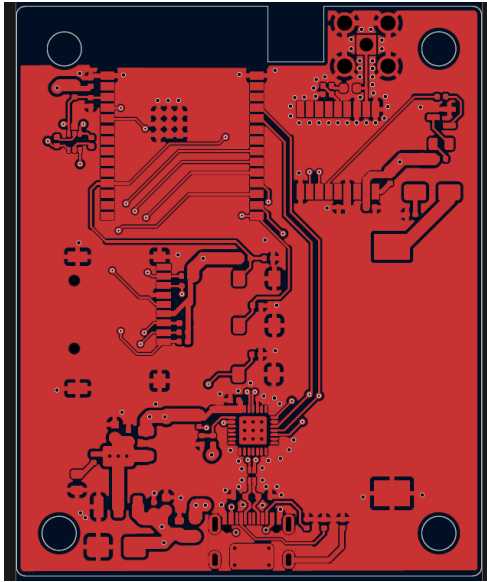
Mileage Logger Project

PCB Layout

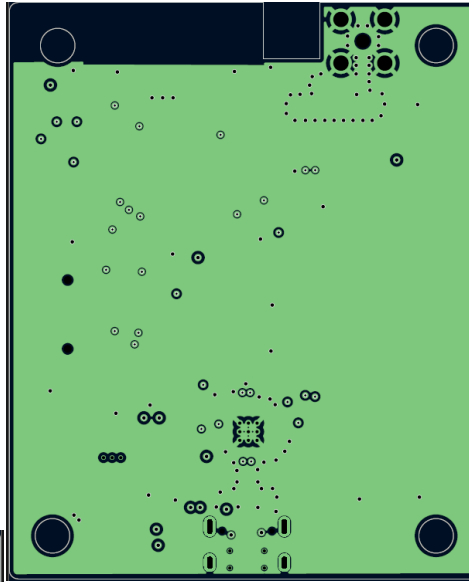


Mileage Logger Project

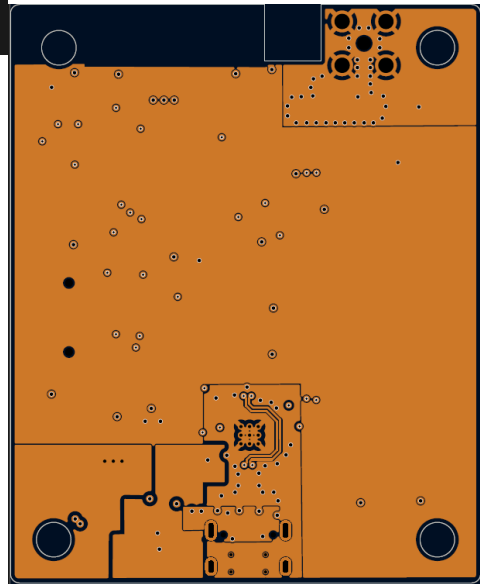
PCB Layer View



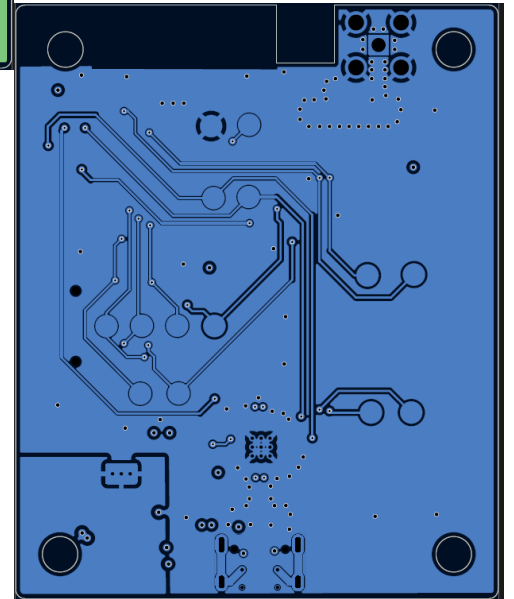
Layer 1 - Signal
(with GND pour)



Layer 2 - GND



Layer 3 - Power
(with GND pour
around differential
pair)



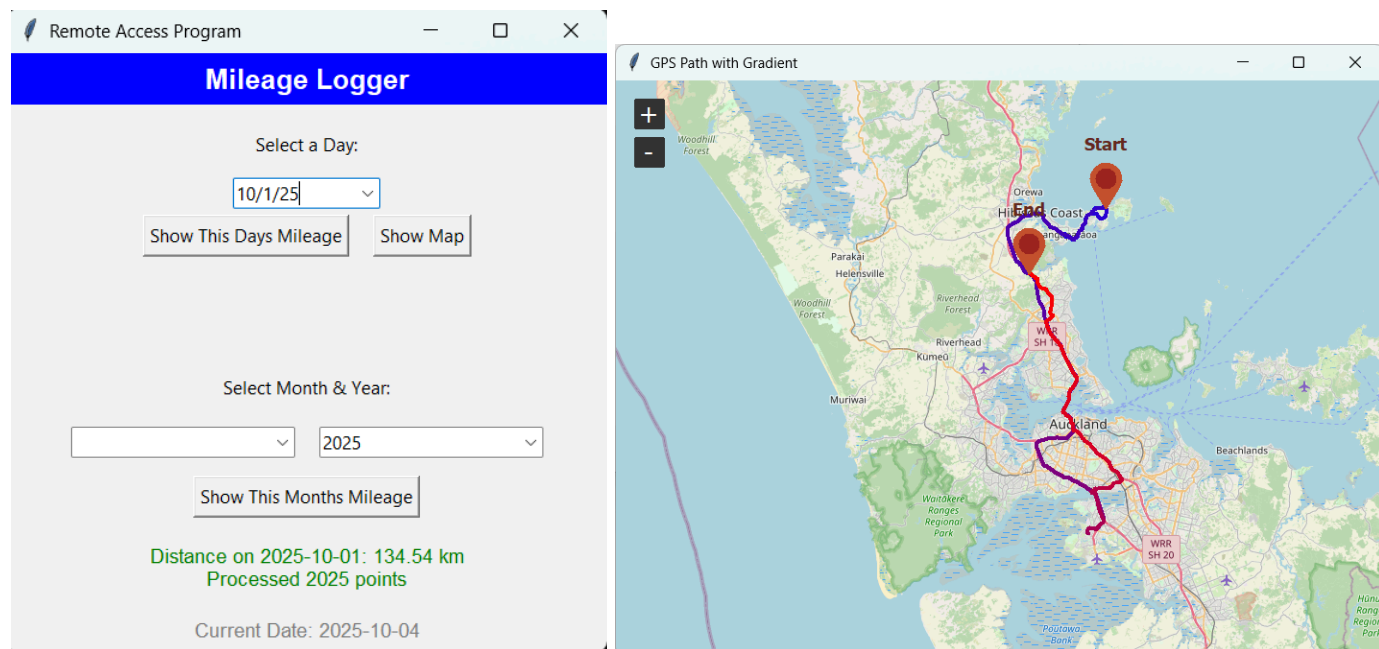
Layer 4 - Signal
(with GND pour)

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Mileage Logger Project

GUI

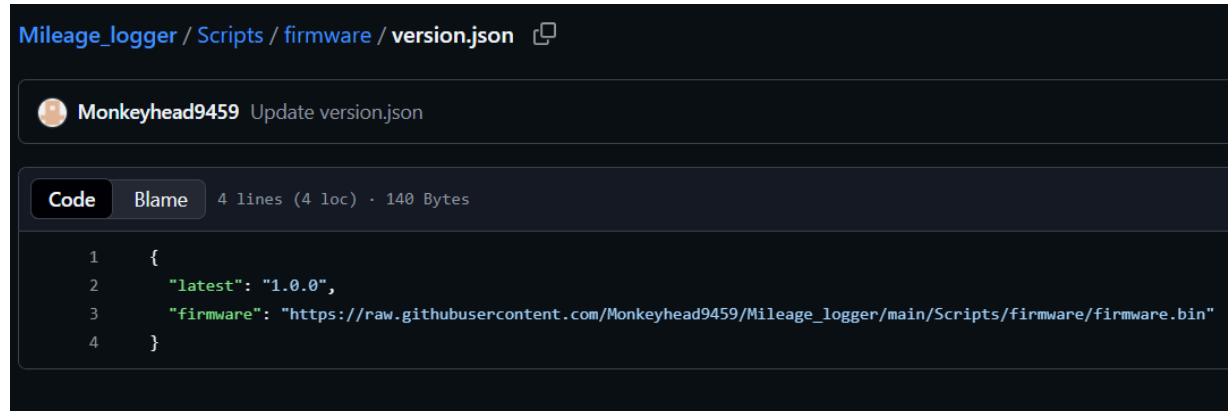
Option 1 - Computer Display



Mileage Logger Project

Back End

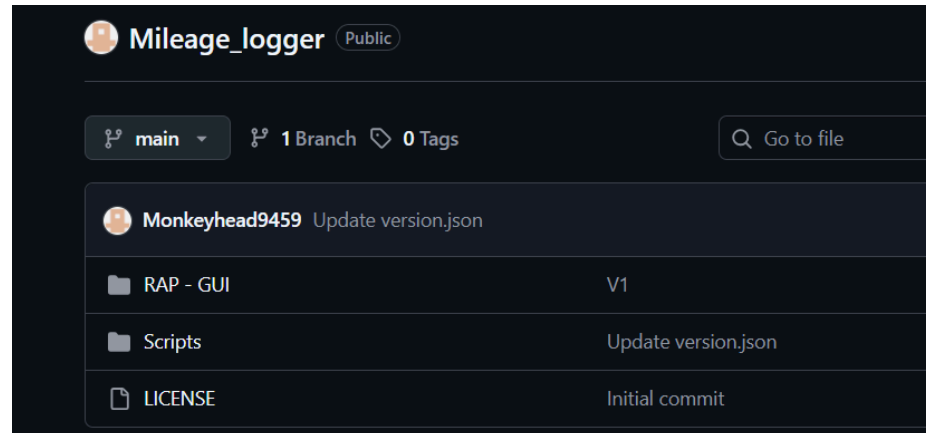
ESP32 REMOTE FIRMWARE UPGRADE THROUGH GITHUB



A screenshot of a GitHub file view for the file `version.json` located at `Mileage_logger / Scripts / firmware / version.json`. The file was updated by user `Monkeyhead9459`. The code is displayed in a dark-themed editor with line numbers 1 through 4. It shows a JSON object with two properties: `latest` and `firmware`. The `latest` property is set to `"1.0.0"`, and the `firmware` property is set to a raw.githubusercontent.com URL pointing to `firmware.bin`.

```
1  {  
2    "latest": "1.0.0",  
3    "firmware": "https://raw.githubusercontent.com/Monkeyhead9459/Mileage_logger/main/Scripts/firmware/firmware.bin"  
4  }
```

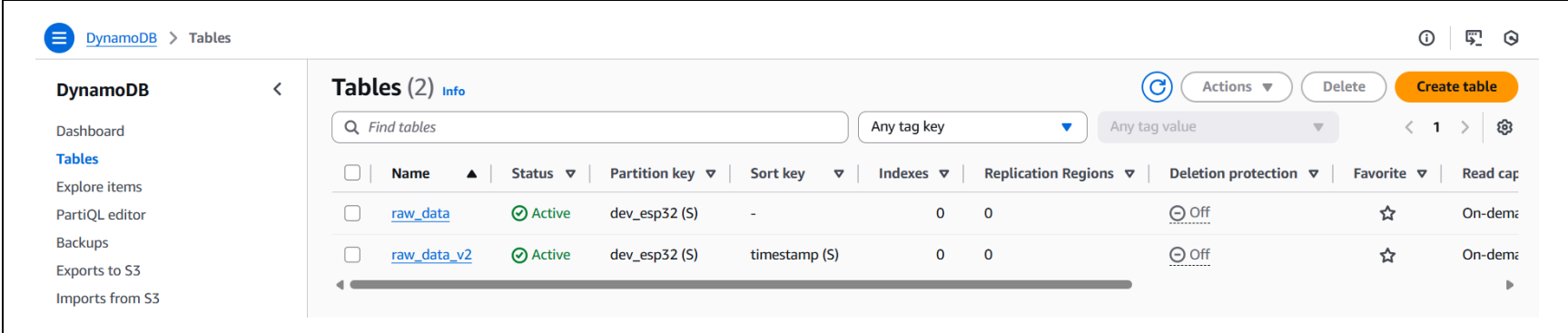
GITHUB VERSION CONTROL



Mileage Logger Project

Back End

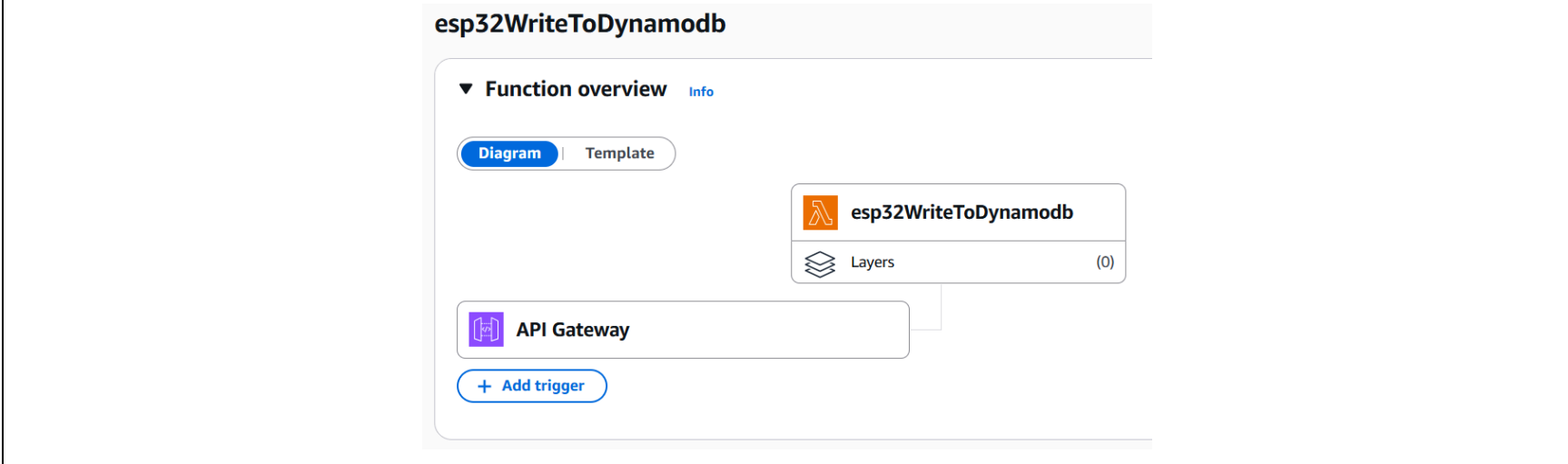
AWS DYNAMODB TABLE FOR CLOUD STORAGE



The screenshot shows the AWS DynamoDB console interface. On the left is a navigation menu with options like Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, and Imports from S3. The main area is titled 'Tables (2)' and contains a table with the following data:

	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read cap
☐	raw_data	Active	dev_esp32 (S)	-	0	0	Off	☆	On-demand
☐	raw_data_v2	Active	dev_esp32 (S)	timestamp (S)	0	0	Off	☆	On-demand

API GATEWAY TO LAMDA FUNCTION FOR ESP32 DATA STORAGE



The screenshot shows the AWS Lambda console for the function 'esp32WriteToDynamodb'. It displays the 'Function overview' tab, which includes a 'Diagram' button, a visual representation of the function with its layers and API Gateway trigger, and an '+ Add trigger' button.

Mileage Logger Project

Improvements:

- Incorporate Google Roads API for more accurate mileage logging by snapping to roads
- Make a portable option for battery connection.
 - SMPS (Increase efficiency)
 - Smaller profile
 - ESP32-C3-MINI-1U
 - Offboard wifi and bluetooth antenna
 - Local/Smaller GPS antenna
- Option 2 for display is making an app through Flutter to connect to the ESP32 through bluetooth and store and display results as well as change journey status from business to personal.